```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=2)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=2)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=2)
)
Total trainable parameters: 360
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:08<00:00, 225.45epoch/s, Loss=2.97e-05, LR=1.0e-03, Best=2.97e-05]
Training finished. Best loss: 2.11e-05. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.692379474639893s
Relative loss: 0.004333922639489174
L2 loss: 0.00275768106803298
Parameter containing:
tensor([[[ 0.1714,  0.7154,  0.8580],
         [-0.0604, -0.3714, -0.7161],
         [-0.2375, -0.6393, -0.1620],
         [ 0.3486,  0.6156,  0.2475],
         [ 0.0658, -0.5223, -0.3984],
         [ 0.2710,  0.8776,  0.8009],
         [-0.0019,  0.6249,  0.5837],
         [-0.0178,  0.3859,  0.6617],
         [ 0.3048,  0.0412, -1.1617],
         [ 0.0706, -0.8446, -0.5525]]], device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[-0.0020, -0.2900, -0.0897],
         [-0.0293,  0.0809, -0.1257],
         [-0.0060, -0.4606,  0.0673],
         [-0.0303, -0.3783,  0.1286],
         [-0.0420,  0.3598,  0.2526],
         [ 0.0702, -0.3527, -0.0505],
         [ 0.0808,  0.5389, -0.4050],
         [-0.0268,  0.2695,  0.0305],
         [ 0.0010, -0.3350, -0.1219],
         [ 0.0208,  0.3543, -0.0420]],

        [[ 0.0816,  0.1998, -0.0582],
         [ 0.0400,  0.0222, -0.0498],
         [-0.0260,  0.1111,  0.1055],
         [-0.0639,  0.2435,  0.1033],
         [-0.0037, -0.3502,  0.1436],
         [-0.0604,  0.2562, -0.0781],
         [ 0.0353, -0.3243, -0.3264],
         [ 0.0275, -0.1242,  0.0030],
         [ 0.0555,  0.1723, -0.0124],
         [ 0.0234, -0.1097,  0.0075]],

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       [[ 0.0574, -0.0428, -0.0674],
        [ 0.0008, -0.0756, -0.0103],
        [-0.0381,  0.0830, -0.0068],
        [-0.0880,  0.2721, -0.0266],
        [-0.1257,  0.0671,  0.0976],
        [ 0.0144,  0.0170, -0.0891],
        [ 0.0560, -0.4149,  0.0182],
        [ 0.0422, -0.0412,  0.0167],
        [ 0.0885, -0.1066, -0.0336],
        [-0.0182, -0.0136, -0.0296]],

       [[ 0.0362,  0.0631, -0.0758],
        [ 0.0273, -0.0092, -0.0219],
        [-0.0434, -0.0949,  0.0270],
        [-0.0278, -0.2662, -0.0901],
        [-0.0798,  0.0147,  0.2050],
        [ 0.0458, -0.1144, -0.1583],
        [ 0.0669,  0.4524,  0.1203],
        [-0.0282,  0.0547,  0.0668],
        [-0.0039,  0.0806, -0.0589],
        [-0.0328,  0.0924, -0.0167]],

       [[-0.0055,  0.4592, -0.0618],
        [ 0.0087, -0.3096, -0.0132],
        [-0.0300,  0.5147,  0.0522],
        [-0.0519,  0.5514,  0.1408],
        [-0.1149, -0.6342,  0.0955],
        [ 0.0612,  0.3621, -0.0854],
        [ 0.0646, -0.7542, -0.2499],
        [ 0.0262, -0.4877, -0.0509],
        [ 0.0785,  0.5276, -0.0968],
        [ 0.0042, -0.5551, -0.0391]],

       [[ 0.0542, -0.5433, -0.1173],
        [-0.0145,  0.6021, -0.0717],
        [-0.0549, -0.7510,  0.1890],
        [-0.1202, -0.4959,  0.0912],
        [-0.0484,  0.4776,  0.2276],
        [ 0.0373, -0.4153, -0.0990],
        [ 0.1281,  0.7570, -0.3962],
        [ 0.0140,  0.6046,  0.0415],
        [ 0.0013, -0.6714, -0.1495],
        [ 0.0164,  0.5433,  0.0195]],

       [[ 0.0311, -0.1264, -0.0781],
        [ 0.0711,  0.0250, -0.0382],
        [ 0.0164, -0.2671,  0.0993],
        [-0.0707, -0.2949,  0.0634],
        [-0.0848,  0.4295,  0.0981],
        [ 0.0046, -0.1548, -0.0677],
        [ 0.1168,  0.7370, -0.2868],
        [-0.0236,  0.2243, -0.0724],
        [ 0.0634, -0.3052, -0.0320],
```

```
         [ 0.0077,  0.1561,  0.0265]],

        [[ 0.0840, -0.4402, -0.0336],
         [ 0.0153, -0.0493, -0.0124],
         [-0.0363, -0.3167,  0.1376],
         [ 0.0088, -0.3676,  0.0750],
         [-0.0570,  0.5851,  0.1633],
         [-0.0184, -0.4025, -0.0046],
         [ 0.0837,  0.4712, -0.2998],
         [-0.0070,  0.4173, -0.0014],
         [ 0.0573, -0.4728, -0.0142],
         [ 0.0104,  0.3790, -0.0035]],

        [[ 0.0767,  0.0747,  0.0306],
         [ 0.0211,  0.0729, -0.0647],
         [-0.0338,  0.0539,  0.1156],
         [-0.0309,  0.1820,  0.2421],
         [-0.0964, -0.3192, -0.1289],
         [ 0.0383,  0.2937,  0.0639],
         [ 0.0656, -0.1193, -0.2347],
         [ 0.0322, -0.1107, -0.0693],
         [ 0.0417,  0.1703, -0.0412],
         [ 0.0194, -0.1114, -0.1258]],

        [[ 0.0731,  0.1992, -0.0321],
         [ 0.0179, -0.0038, -0.0583],
         [-0.0503,  0.2676,  0.1525],
         [ 0.0264,  0.3755,  0.2190],
         [-0.0774, -0.3495,  0.1941],
         [-0.0276,  0.3682, -0.0548],
         [ 0.0563, -0.4719, -0.4253],
         [-0.0125, -0.1804,  0.0114],
         [ 0.0214,  0.2326, -0.0527],
         [ 0.0173, -0.2999, -0.0539]]], device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[ 0.0156,  0.0774, -0.0029]],

        [[ 0.0262,  0.1673,  0.0094]],

        [[ 0.0490, -0.1210, -0.0448]],

        [[ 0.0327, -0.1025, -0.1400]],

        [[-0.0145, -0.1830, -0.0767]],

        [[ 0.0236,  0.1011, -0.1183]],

        [[-0.0247,  0.0889, -0.0920]],

        [[ 0.0159,  0.0116, -0.1189]],

        [[ 0.0682,  0.0874, -0.0205]],

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        [[ 0.0250,  0.0655, -0.1506]]], device='cuda:0', requires_grad=True)
=====================================================================================
======
Layer (type:depth-idx)                   Output Shape              Param #
=====================================================================================
======
CPIKANModel                              [1, 1]                    --
├─ModuleList: 1-1                         --                        --
│    └─0.weight                                                      ├─30
│    └─1.weight                                                      ├─300
│    └─2.weight                                                      └─30
│    └─ChebyKANLayer: 2-1                  [1, 10]                 30
│    │    └─weight                                                   └─30
│    └─ChebyKANLayer: 2-2                  [1, 10]                 300
│    │    └─weight                                                   └─300
│    └─ChebyKANLayer: 2-3                  [1, 1]                  30
│    │    └─weight                                                   └─30
=====================================================================================
======
Total params: 360
Trainable params: 360
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
=====================================================================================
======
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
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进程已结束，退出代码为 0

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